

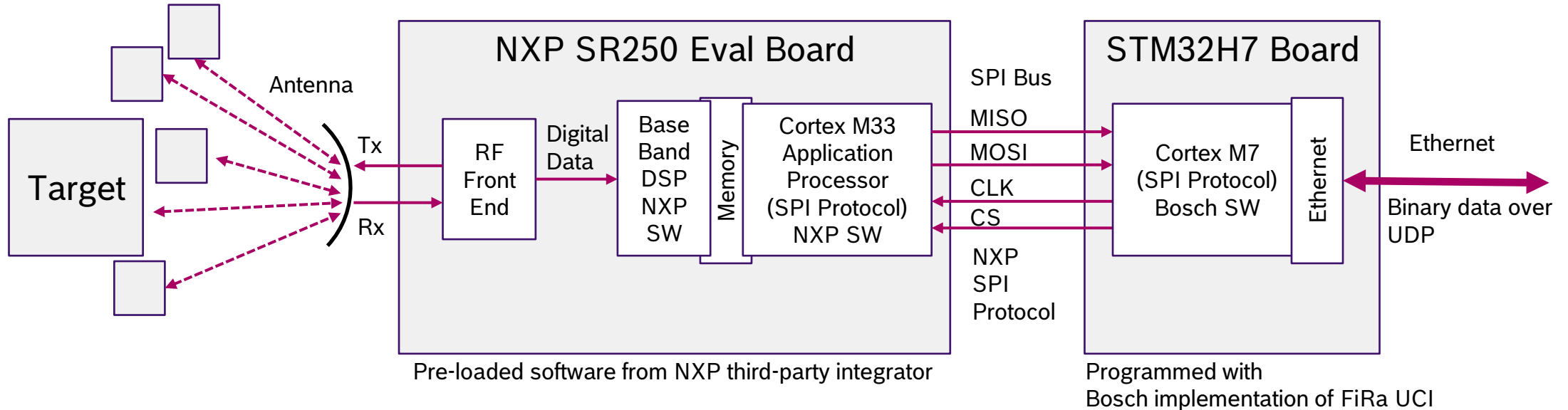


Bosch UWB Measurement Setup

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Bosch UWB Measurement Setup

Hardware Setup Overview



FiRa: Fine Ranging Consortium [<https://www.firaconsortium.org/>]

UCI: UWB Command Interface

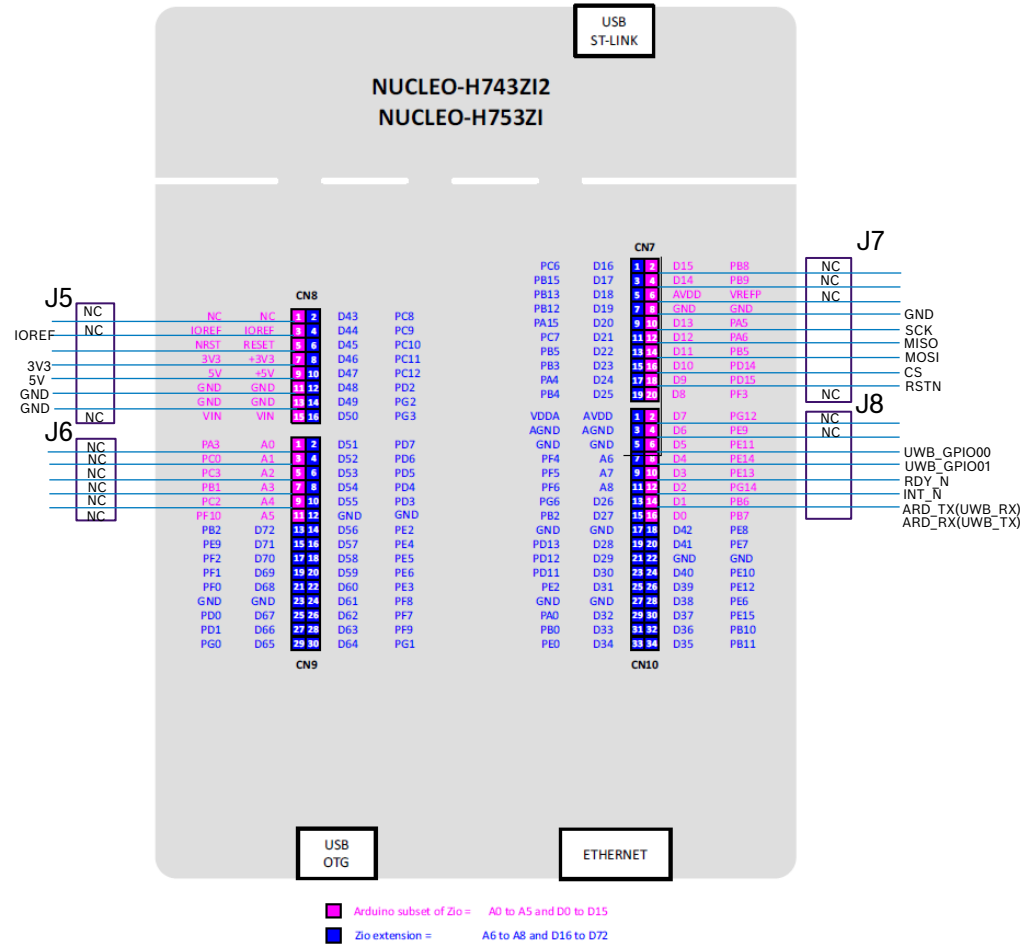
Internal C-SC1 | CR/RCS1-NA | 2026-02-27

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Pinout Maps

Figure 18. NUCLEO-H743ZI2 and NUCLEO-H753ZI2



Nucleo	TS250
PA5	SCK
PA6	MISO
PB5	MOSI
PD14	CS
PD15	RST_N
PE11	UWB_GPIO00
PE14	UWB_GPIO01
PE13	RDY_N
PG14	INT_N
PB6	ARD_TX(UWB_RX)
PB7	ARD_RX(UWB_TX)

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Software and Documentation Pre-requisites

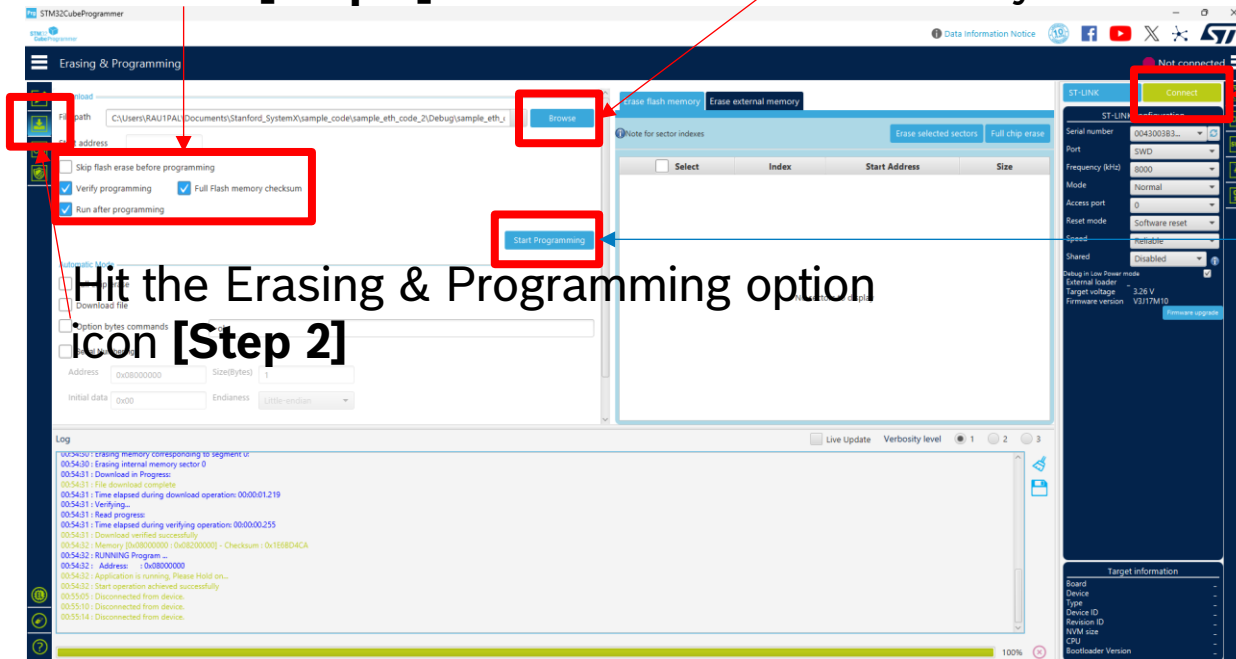
- Please install STM32CubeProgrammer software [[Link](#)]
- Allows update of firmware on boards to add/modify/remove functions

Ensure the settings are as in the screenshot **[Step 4]**

Hit the Browse button and select the .elf binary file to be flashed **[Step 3]**

After connecting the board over USB-mini cable, hit Connect **[Step 1]**

Hit Start Programming button **[Step 5]**



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Ethernet Setup

- STM32H7 board is pre-configured to be on IP 192.168.1.10
 - Listening on port 37249
- Device connecting to STM32H7 over ethernet should be on IP 192.168.1.102
 - Listening on port 20000
- DHCP is best set to Manual on the device with IP 192.168.1.102

 Ethernet
Not connected 

Authentication settings

Edit

Metered connection

Some apps might work differently to reduce data usage when you're connected to this network

Off 

Set a data limit to help control data usage on this network

IP assignment:

Manual

IPv4 address:

192.168.1.102

IPv4 mask:

255.255.255.0

Edit

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Ethernet Input Structure

```
typedef struct {  
    uint8_t cmd_id; // 1: Set radar configuration index and duration of run  
    uint8_t setting_idx;  
    uint32_t duration_in_ms;  
}udp_input_cmd_t;
```

- cmd_id → 1
 - Send radar configuration index and duration of run command

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Ethernet Output Structure

Capturing from Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1038	727.779153	192.168.1.102	224.0.0.251	MDNS	76	Standard query 0x0000 A S10VW05860.local, "QM" question
1039	727.779762	fe80::4aea:62ff:fe02::fb	ff02::fb	MDNS	96	Standard query 0x0000 A S10VW05860.local, "QM" question
1040	727.780255	192.168.1.102	224.0.0.251	MDNS	76	Standard query 0x0000 AAAA S10VW05860.local, "QU" question
1041	727.780653	fe80::4aea:62ff:fe02::fb	ff02::fb	MDNS	96	Standard query 0x0000 AAAA S10VW05860.local, "QU" question
1042	728.202233	192.168.1.102	224.0.0.251	MDNS	76	Standard query 0x0000 A S10VW05860.local, "QM" question
1043	728.202625	192.168.1.102	192.168.1.255	NBNS	92	Name query NB S10VW05860<0>
1044	728.202735	fe80::4aea:62ff:fe02::fb	ff02::fb	MDNS	96	Standard query 0x0000 A S10VW05860.local, "QM" question
1045	728.203078	192.168.1.102	224.0.0.251	MDNS	76	Standard query 0x0000 AAAA S10VW05860.local, "QU" question
1046	728.203384	fe80::4aea:62ff:fe02::fb	ff02::fb	MDNS	96	Standard query 0x0000 AAAA S10VW05860.local, "QU" question
1047	728.625986	192.168.1.102	224.0.0.251	MDNS	76	Standard query 0x0000 AAAA S10VW05860.local, "QU" question
1048	728.626474	fe80::4aea:62ff:fe02::fb	ff02::fb	MDNS	96	Standard query 0x0000 AAAA S10VW05860.local, "QU" question
1049	728.626801	192.168.1.102	224.0.0.251	MDNS	76	Standard query 0x0000 A S10VW05860.local, "QM" question
1050	728.627404	fe80::4aea:62ff:fe02::fb	ff02::fb	MDNS	96	Standard query 0x0000 A S10VW05860.local, "QM" question
1051	728.955676	192.168.1.102	192.168.1.255	NBNS	92	Name query NB S10VW05860<0>
1052	729.708342	192.168.1.102	192.168.1.255	NBNS	92	Name query NB S10VW05860<0>
1053	730.487631	192.168.1.102	192.168.1.255	NBNS	92	Name query NB S10VW05860<0>
1054	731.248790	192.168.1.102	192.168.1.255	NBNS	92	Name query NB S10VW05860<0>
1055	732.004207	192.168.1.102	192.168.1.255	NBNS	92	Name query NB S10VW05860<0>
1056	733.469662	192.168.1.10	192.168.1.102	IPv4	1514	Fragmented IP protocol (proto=UDP 17, off=0, ID=0001)
1057	733.469753	192.168.1.10	192.168.1.102	IPv4	1514	Fragmented IP protocol (proto=UDP 17, off=1480, ID=0001) [Reassembled in #1058]
1058	733.469852	192.168.1.10	192.168.1.102	UDP	1188	62510 → 20000 Len=4106
1059	737.153250	192.168.1.102	224.0.0.251	MDNS	76	Standard query 0x0000 A s10vm05750.local, "QM" question
1060	737.153764	fe80::4aea:62ff:fe02::fb	ff02::fb	MDNS	96	Standard query 0x0000 A s10vm05750.local, "QM" question
1061	737.565933	192.168.1.102	224.0.0.251	MDNS	76	Standard query 0x0000 A s10vm05750.local, "QM" question

> Frame 1058: Packet, 1188 bytes on wire (9504 bits), 1188 bytes captured (9504 bits) on interface \Device\NPF...

> Ethernet II, Src: STHMicroelect_00:00:01 (00:80:e1:00:00:01), Dst: HP_f1:07:7e (48:ea:62:f1:07:7e)

> Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.1.102

> User Datagram Protocol, Src Port: 62510, Dst Port: 20000

> Data (4106 bytes)

[Data ...]

0000 f4 2e 4e 20 10 12 00 00 01 00 00 f0 00 00 08 00

0010 80 00 05 00 00 00 21 f4 c4 00 13 08 32 00 e0 b8

0020 c4 00 01 61 01 01 08 05 02 04 21 3c 5b 00 00 00

0030 77 02 fd 00 f6 00 cc 07 2e 06 4b 0f 5b 0a 97 f5

0040 32 0c d7 c3 4f 31 66 d0 92 65 8f 15 bd 54 2e 2b

0050 5f 0e ef 06 0d f3 ee f5 8a 05 56 06 ea 08 42 0c

0060 35 f8 04 ff 90 f4 80 fa 2e fe 6b 03 d4 fe 9e 04

0070 0f f6 0c fc 1d f6 b6 fb 5c fd 94 00 ec fb cf fd

0080 18 f9 f4 fb 18 fe ca fe 47 00 e4 fe 4b ff b4 fe

0090 3b 00 91 00 1f 00 1f 01 d1 fe 18 00 f6 fe dd ff

00a0 fd ff 65 00 19 00 73 00 4c ff d8 ff 0e ff 75 ff

00b0 f0 ff f9 ff 9c 00 9e 00 30 00 8a 00 d3 ff 70 00

00c0 44 00 bf 00 65 00 7b 00 bb ff e5 ff a6 ff e9 ff

00d0 21 00 08 00 41 00 01 00 2e 00 1b 00 05 0e fe ff

00e0 c9 ff bd ff ff d3 ff 5c 00 1f 00 71 00 34 00

00f0 34 00 12 00 00 e8 ff 00 cd ff f7 ff c3 ff

0100 db ff b8 ff e0 ff c8 ff 06 00 f3 ff 09 06 fe ff

0110 ea ff e0 ff e2 ff ea ff ed ff 01 00 f8 ff 11 00

0120 fa ff 25 00 f8 ff 26 00 f5 ff 13 00 0e 00 1c 00

0130 24 00 14 00 03 00 12 00 f7 ff 0f 00 05 00 05 00

0140 02 00 00 00 03 00 f0 ff 03 00 ed ff 03 00 f1 ff

0150 0e 00 05 00 0b 00 0a 00 ff ff 0c 00 f8 ff 03 00

0160 fb ff 14 00 06 00 28 00 e6 ff 0a 00 d6 ff ea ff

0170 ee ff ec ff 03 00 f6 ff 02 00 f4 ff fd ff f9 ff

0180 0d 00 fa ff 01 00 f1 ff 01 00 fc ff 13 00 0e 00

0190 12 00 07 00 01 00 f4 ff fc ff f3 ff 03 00 0e 00

01a0 05 00 11 00 f4 ff f7 ff f0 ff f8 ff fd ff 01 00

01b0 02 00 03 00 ff ff fb ff fc ff f5 ff 06 00 fc ff

Packet (1188 bytes) Reassembled IPv4 (4114 bytes) Packets: 1115

Data (data.data), 4,106 bytes

0000	f4 2e 4e 20 10 12 00 00	01 00 00 f0 00 00 08 00
0010	80 00 05 00 00 00 21 f4	c4 00 13 08 32 00 e0 b8
0020	c4 00 01 61 01 01 08 05	02 04 21 3c 5b 00 00 00
0030	77 02 fd 00 f6 00 cc 07	2e 06 4b 0f 5b 0a 97 f5
0040	32 0c d7 c3 4f 31 66 d0	92 65 8f 15 bd 54 2e 2b
0050	5f 0e ef 06 0d f3 ee f5	8a 05 56 06 ea 08 42 0c
0060	35 f8 04 ff 90 f4 80 fa	2e fe 6b 03 d4 fe 9e 04
0070	0f f6 0c fc 1d f6 b6 fb	5c fd 94 00 ec fb cf fd
0080	18 f9 f4 fb 18 fe ca fe	47 00 e4 fe 4b ff b4 fe
0090	3b 00 91 00 1f 00 1f 01	d1 fe 18 00 f6 fe dd ff
00a0	fd ff 65 00 19 00 73 00	4c ff d8 ff 0e ff 75 ff
00b0	f0 ff f9 ff 9c 00 9e 00	30 00 8a 00 d3 ff 70 00
00c0	44 00 bf 00 65 00 7b 00	bb ff e5 ff a6 ff e9 ff
00d0	21 00 08 00 41 00 01 00	2e 00 1b 00 05 00 fe ff
00e0	c9 ff bd ff ff d3 ff	5c 00 1f 00 71 00 34 00
00f0	34 00 12 00 00 e8 ff	00 cd ff f7 ff c3 ff
0100	db ff b8 ff e0 ff c8 ff	06 00 f3 ff 09 00 f6 ff
0110	ea ff e0 ff e2 ff ea ff	ed ff 01 00 f8 ff 11 00
0120	fa ff 25 00 f8 ff 26 00	f5 ff 13 00 0e 00 1c 00
0130	24 00 14 00 03 00 12 00	f7 ff 0f 00 05 00 05 00
0140	02 00 00 00 03 00 f0 ff	03 00 ed ff 03 00 f1 ff
0150	0e 00 05 00 0b 00 0a 00	ff ff 0c 00 f8 ff 03 00
0160	fb ff 14 00 06 00 28 00	e6 ff 0a 00 d6 ff ea ff
0170	ee ff ec ff 03 00 f6 ff	02 00 f4 ff fd ff f9 ff
0180	0d 00 fa ff 01 00 f1 ff	01 00 fc ff 13 00 0e 00
0190	12 00 07 00 01 00 f4 ff	fc ff f3 ff 03 00 0e 00
01a0	05 00 11 00 f4 ff f7 ff	f0 ff f8 ff fd ff 01 00
01b0	02 00 03 00 ff ff fb ff	fc ff f5 ff 06 00 fc ff



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Ethernet Output Structure

- Advise received from the integrators to break long ethernet CIR report packets into smaller chunks
- Decision taken to send chunks of 1184 bytes of raw CIR information in each ethernet packet
- Structure of CIR information packet is as follows

```
typedef struct {  
    uint8_t cmd_id; // 2: CIR report fragment  
    uint8_t last_fragment; // 0 --> no, 1 --> yes  
    uint16_t data_len;  
    uint8_t data[MAX_DATA_PER_ETH_PACKET];  
}udp_cir_fragment_t;
```

- MAX_DATA_PER_ETH_PACKET is 1184
- Pseudocode of reception and re-assembly of raw CIR information
 - Instantiate a raw CIR information buffer capable of holding at most 10250* bytes
 - Do the following if received ethernet packet cmd_id → 2
 - If last_fragment == 0 → append *data_len* bytes from *data* to raw CIR information buffer
 - If last_fragment == 1 → append *data_len* bytes from *data* to raw CIR information buffer. Buffering is complete. Send upstream raw CIR information buffer for further processing

*RFRI will be set to 10ms. A complete radar activity interval is 100ms. There are at most 2 Rx paths. Each CIR is at most 128 taps. Each tap is 4 bytes. There are additional 10 frame bytes in RADAR_RX_NTF. Total → $(100\text{ms} / 10\text{ms}) * 2 * 128 * 4 \text{ bytes} = 10250 \text{ bytes}$.

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Radar Configuration Decisions

- Ch #5 or Ch #9
 - Only Ch #9 will be tested [**Rationale:** The antennas on-board are specifically tuned for Ch #9 only. Since physical spacing between the antennas is determined by the channel, we cannot use Ch #5]
- Radar modes
 - Only modes 1 and 3 will be used [**Rationale:** 2 is discouraged by the vendor. 4, 5 and 6 in LPRF domain radiate lower power and take a longer duration. Will never be used in any practical use-case scenario.]
- RFRI
 - An RFRI of 10ms will be used for all test runs
 - **Rationale:**
 - Standards discussion suggests 1%-5% to be the range of duty-cycle allowed for UWB radar.
 - For modes 1 and 3, SYNC duration [Table 1 or UWB Radar Fundamentals pdf] is 93us and 186us respectively.
 - SFD duration is negligible, but we can assume the Preamble + SFD duration to be ~100us (for mode 1) and ~195ms (for mode 3)
 - Considering the minimum of the two, a 1% duty-cycle evaluates to 10ms, and 5% duty-cycle evaluates to 20ms RFRI
 - We will consider the minimum of the two RFRI. If longer RFRI (will always be in multiple of the minimum RFRI) is needed we will drop data during post-process

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Radar Configuration Decisions

- AGC (Automatic Gain Control) of the Rx Antenna
 - Both AGC active and manual gain settings will be tested
- Tx power
 - We will be testing two power levels. A default one for mode #1 and half of that for mode #3
- Off-board antenna placement
 - Placed at equivalent distance between RXC and RXB to the right of RXB in the same horizontal line. All references are from the center of each antenna